

Enhancing Clinical Decision-Making in Heart Transplantation Through Predictive Modeling

Erik Pak

Advisor: Dr. Ilyas Ustun

DePaul University

erikpak.email@gmail.com

ABSTRACT

Heart transplantation is a life-saving intervention for patients with end-stage heart failure, yet predicting long-term survival remains a complex challenge due to the multifaceted nature of donor and recipient characteristics. This study leverages the United Network for Organ Sharing (UNOS) dataset—a large-scale, nationally representative registry of heart transplant cases, which is a crucial resource in the field of heart transplantation research. The UNOS dataset provides comprehensive and reliable data on donor and recipient characteristics, making it an ideal source for developing and evaluating machine learning models for predicting three-year post-transplant survival. Our study follows a comprehensive data science pipeline which includes data acquisition, rigorous preprocessing (imputation,

discretization, and normalization), exploratory data analysis, advanced feature engineering, and feature selection. We benchmarked multiple classification algorithms—Logistic Regression, Random Forest, XGBoost, K-Nearest Neighbors (KNN), and AdaBoost—using stratified cross-validation with Bayesian hyperparameter optimization. Model performance was assessed using AUC-ROC, F1-score, precision, and recall. SHAP (SHapley Additive exPlanations) values were employed to interpret model predictions and identify key predictors of survival, ensuring transparency and robustness in our research. This study aims to improve predictive accuracy and support evidence-based clinical decision-making in heart transplantation by comparing model performance and uncovering the most influential features.

Keywords

Machine Learning Classification Algorithms Feature Engineering Data Preprocessing
Bayesian Optimization Heart Transplant

INTRODUCTION

Since the first human heart transplant on December 3, 1967, by surgeon Christiaan Barnard at Groote Schuur Hospital in Cape Town, South Africa, where a 25-year-old woman fatally injured in a car accident was transplanted into Lewis Washkansky, a 53-year-old South African grocer dying from chronic heart disease. Barnard's achievement was built upon years of pioneering research by American surgeon Norman Shumway and his colleagues at Stanford University, who had developed and refined heart transplantation techniques in animal models throughout the 1950s and achieved the first successful heart transplant in a dog at Stanford University in California in 1958[1].

We focused on adult patients who underwent heart transplantation between 2011 and 2021 to ensure sufficient follow-up data and account for any pharmaceutical and surgical advances for a three-year survival analysis. The

dataset includes various demographic, clinical, procedural, and post-operative variables used to explore advanced predictive analytics in this domain. Traditional statistical methods often struggle to model complex nonlinear relationships and variable interactions. In contrast, modern machine learning approaches can capture these intricate patterns, potentially enhancing clinical decision-making.

In this context, our study addresses the critical challenge of predicting three-year post-transplant survival by applying machine learning algorithms to heart transplant patients from the UNOS database, a comprehensive and reliable source of transplant data. We aim to balance predictive performance with interpretability, enabling clinicians to trust the outputs and understand the underlying factors driving outcomes. This research ultimately seeks to support more informed decision-making in allocating and managing donor hearts.

RecipientStatus	Dead	Living	Lost to Follow Up	Missing	Retransplanted	Row Total	Dead %	Living %
ListingYear								
2011	664.0	1273.0	47.0	0.0	34.0	2018.0	32.903865	63.082260
2012	643.0	1438.0	32.0	0.0	35.0	2148.0	29.934823	66.945996
2013	620.0	1568.0	38.0	0.0	31.0	2257.0	27.470093	69.472751
2014	576.0	1791.0	26.0	0.0	26.0	2419.0	23.811492	74.038859
2015	512.0	1963.0	28.0	0.0	23.0	2526.0	20.269200	77.711797
2016	496.0	2112.0	13.0	0.0	17.0	2638.0	18.802123	80.060652
2017	436.0	2191.0	5.0	0.0	12.0	2644.0	16.490166	82.866868
2018	357.0	2363.0	4.0	3.0	24.0	2751.0	12.977099	85.896038
2019	358.0	2494.0	1.0	12.0	8.0	2873.0	12.460842	86.808214
2020	257.0	2463.0	0.0	16.0	13.0	2749.0	9.348854	89.596217
2021	114.0	1989.0	0.0	359.0	9.0	2471.0	4.613517	80.493727
Column Total	5033.0	21645.0	194.0	390.0	232.0	27494.0	18.305812	78.726268

Figure 1: Contingency Table

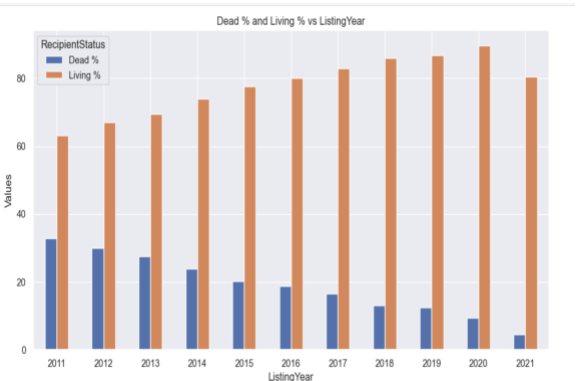


Figure 2: Survival & Death Rate

PREPROCESSING

We focused our analysis on adult patients, defined as individuals aged 18 years or older, who received a heart transplant between 2011 and 2021. This timeframe was selected to ensure the availability of sufficient follow-up data to evaluate three-year post-transplant survival outcomes, which served in the study. By restricting to this period, we aimed to capture contemporary clinical practices and outcomes relevant to current heart transplantation care.

To ensure the quality and relevance of our dataset, we embarked on a meticulous data cleaning process. We began with a comprehensive registry from the United Network for Organ Sharing (UNOS), which initially included data from 1984 to 2021 and removed any variables that were not currently being used or any duplicate variables. After applying inclusion criteria and data

cleaning procedures, the resulting dataset contained 28,751 patient records and 317 features. To address the issue of incomplete data, we excluded features with greater than 80% missing values since they were primarily categorical features; therefore, a new categorical feature called "Missing" was created for any missing features that were part of the inclusion, thereby retaining enough categorical variables for meaningful analysis.

Further data preprocessing steps were undertaken to enhance the dataset's consistency and usability. Categorical features were mapped according to the official UNOS data dictionary, ensuring standardized representation across all variables. Additionally, we updated the data types within the DataFrame to accurately reflect the nature of each variable, such as converting category strings to categorical objects.

VISUALIZATION



Figure 3: Candidate Drug Usage

ZOSYN (piperacillin and tazobactam for injection, USP) is an injectable antibacterial combination product consisting of the semisynthetic antibiotic piperacillin sodium and the β -lactamase inhibitor tazobactam sodium for intravenous administration. Piperacillin sodium is derived from D(-)- α -aminobenzyl-penicillin.

Rocuronium is a non-depolarizing neuromuscular blocker widely used to produce muscle relaxation to help facilitate surgery and ventilation of the lungs in elective and emergent situations.

Vancomycin is a tricyclic glycopeptide antibiotic originally derived from the organism *Streptococcus orientalis*. Vancomycin is used to treat and prevent various bacterial infections caused by gram-positive bacteria, including methicillin-resistant *Staphylococcus aureus* (MRSA).

Vecuronium is a nondepolarizing agent that achieves skeletal muscle paralysis by competing with acetylcholine for cholinergic receptor sites and binding with the nicotinic cholinergic receptor at the postjunctional membrane of the motor endplate

Albuterol is used to prevent and treat difficulty breathing, wheezing, shortness of breath, coughing, and chest tightness caused by lung diseases such as asthma and chronic obstructive pulmonary disease (COPD; a group of diseases that affect the lungs and airways).

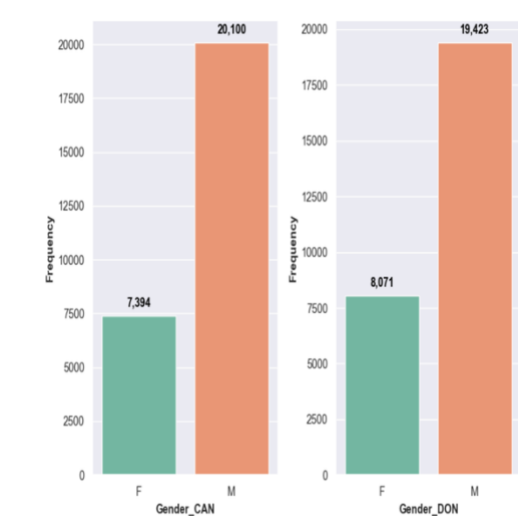
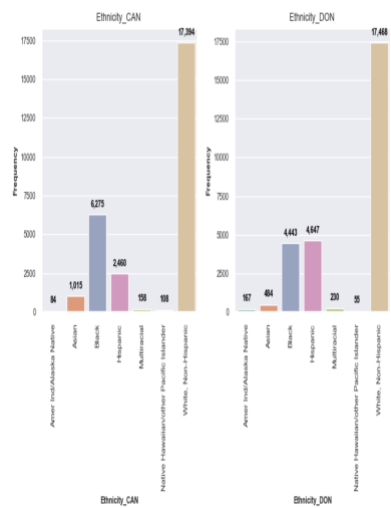


Figure 4: Gender

The gender distribution in Gender_CAN (Candidate) and Gender_DON (Donor) is similar, with about 70% males and 30% females, slightly more in female donors. This indicates a consistent demographic pattern across candidates and donors, suggesting that males have a higher frequency of heart transplants in the UNOS dataset.

RecipientStatus	Dead	Living	Lost to Follow Up	Missing	Retransplanted	Row Total	Dead %	Living %
Gender_CAN								
F	1345.0	5811.0	51.0	98.0	89.0	7394.0	18.190425	78.590749
M	3688.0	15834.0	143.0	292.0	143.0	20100.0	18.348259	78.776119
Column Total	5033.0	21645.0	194.0	390.0	232.0	27494.0	18.305812	78.726268



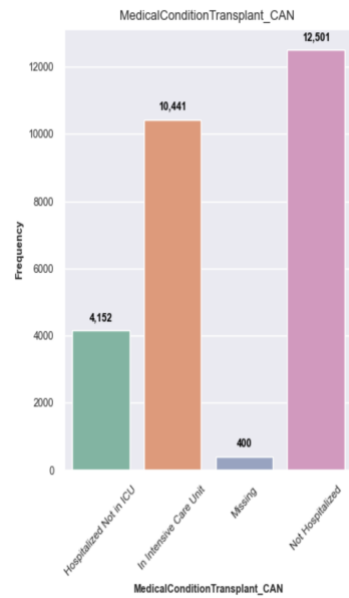
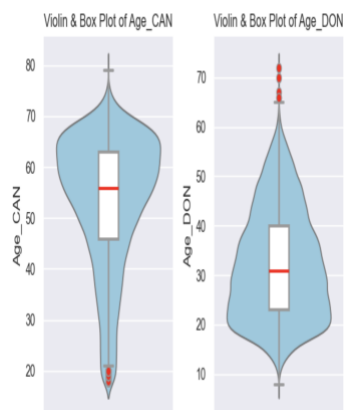
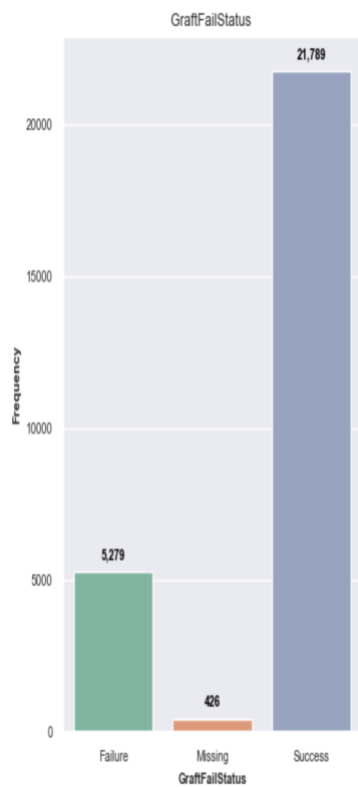
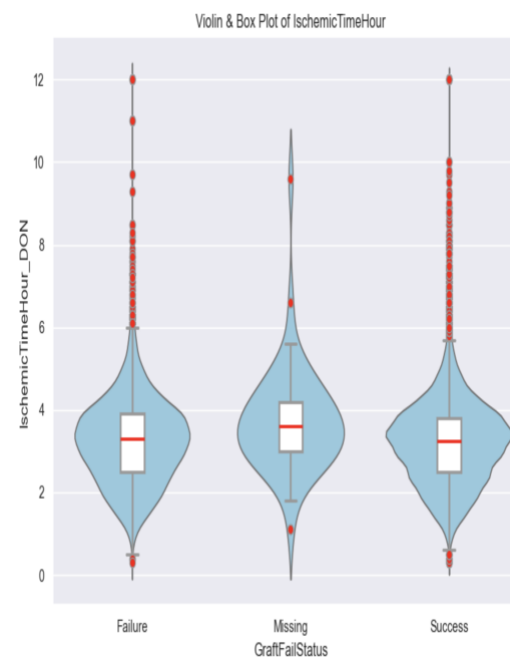
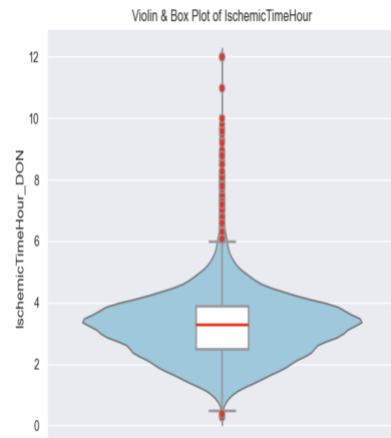
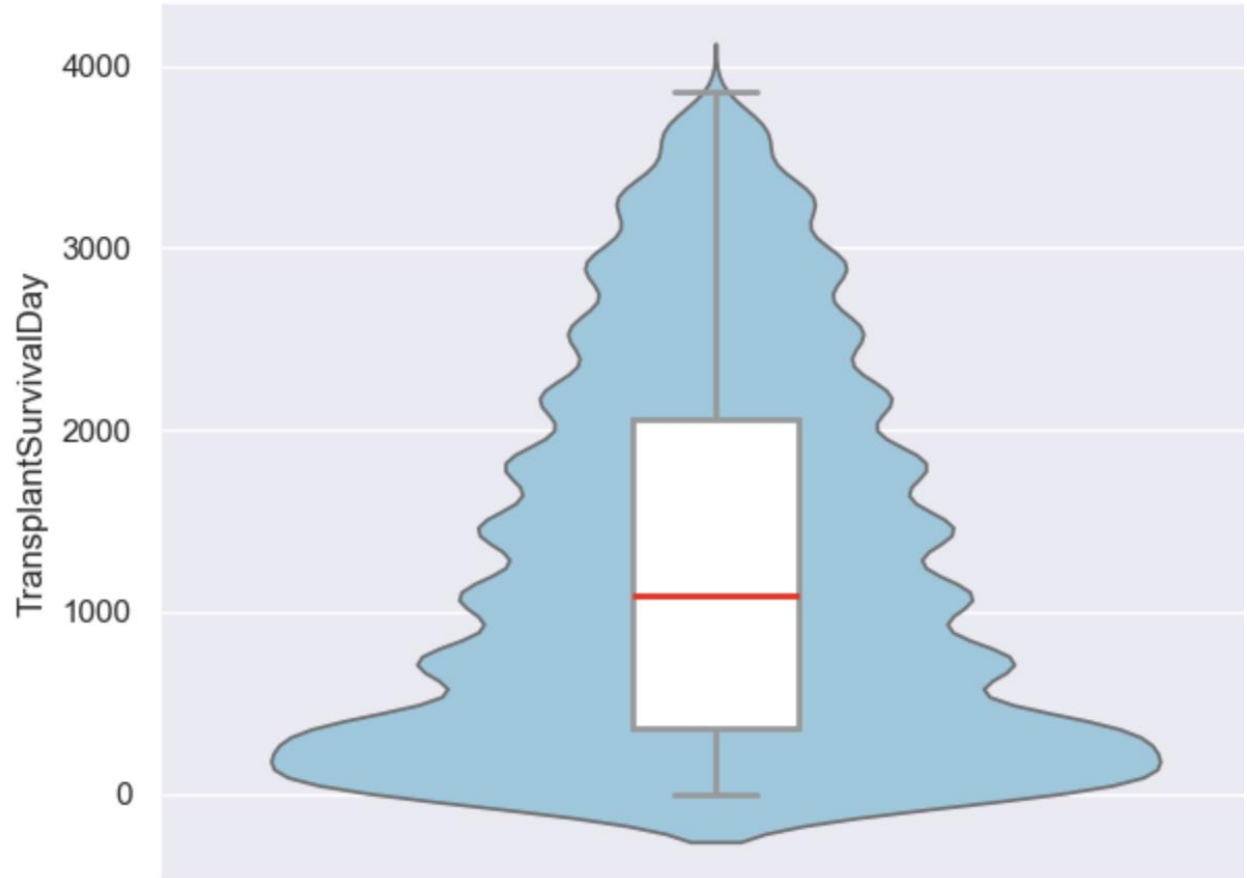


Figure 5: Functional Status at Transplant

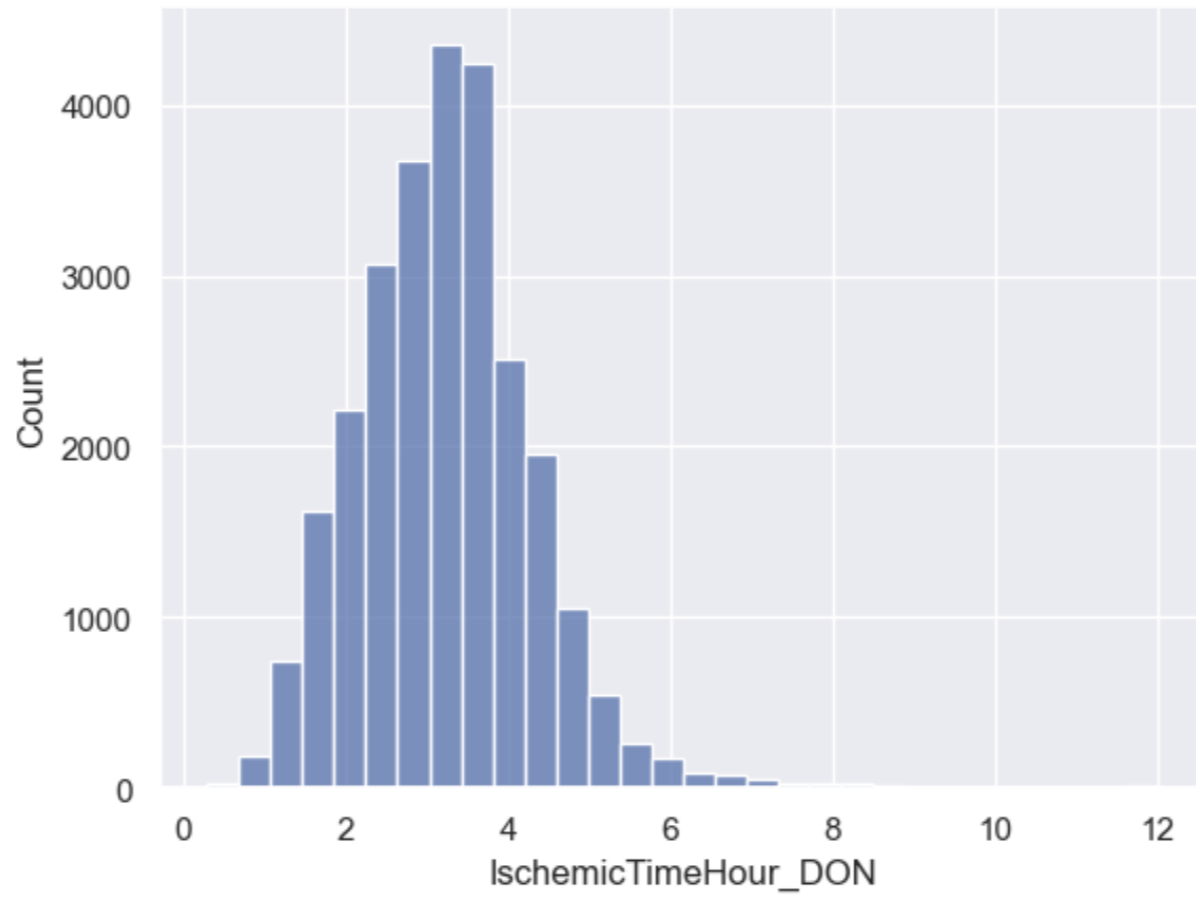




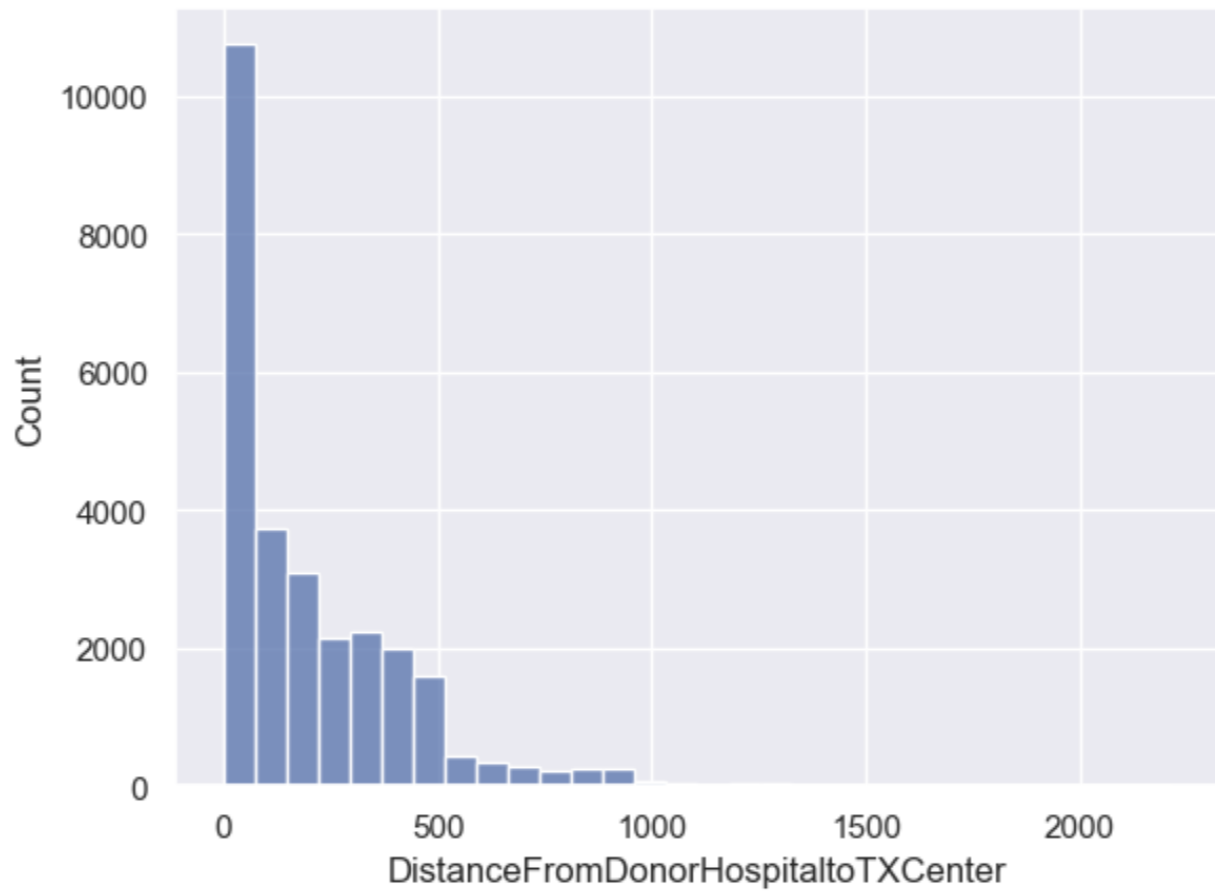
Violin & Box Plot of Transplant Survival Day(s)

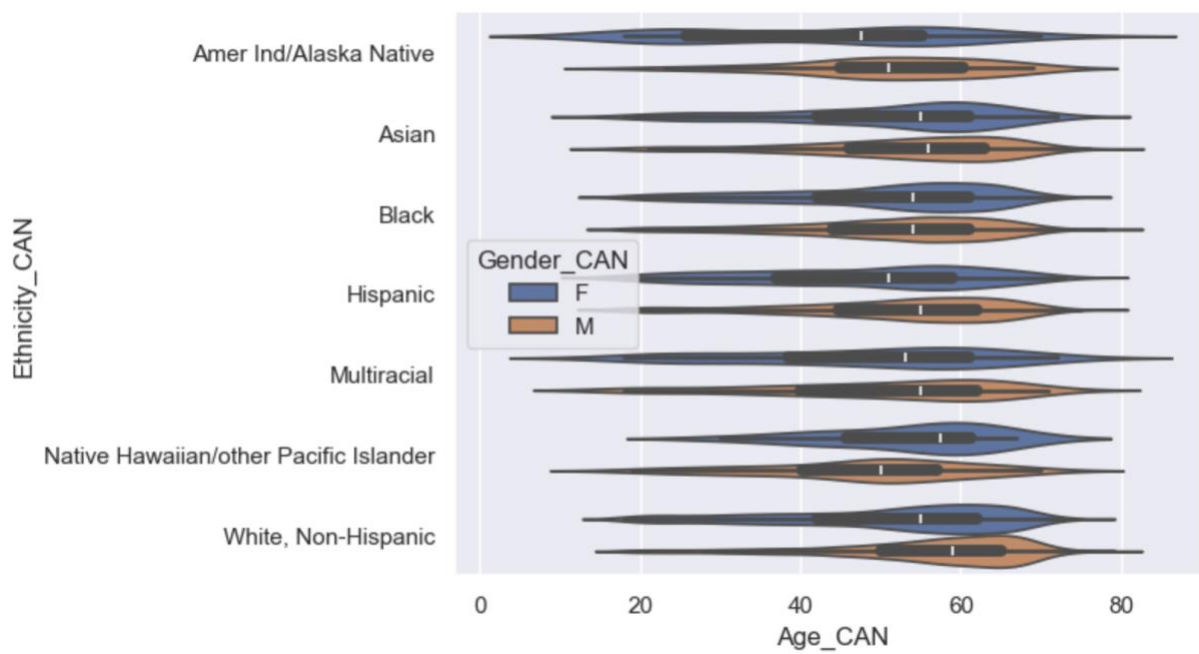
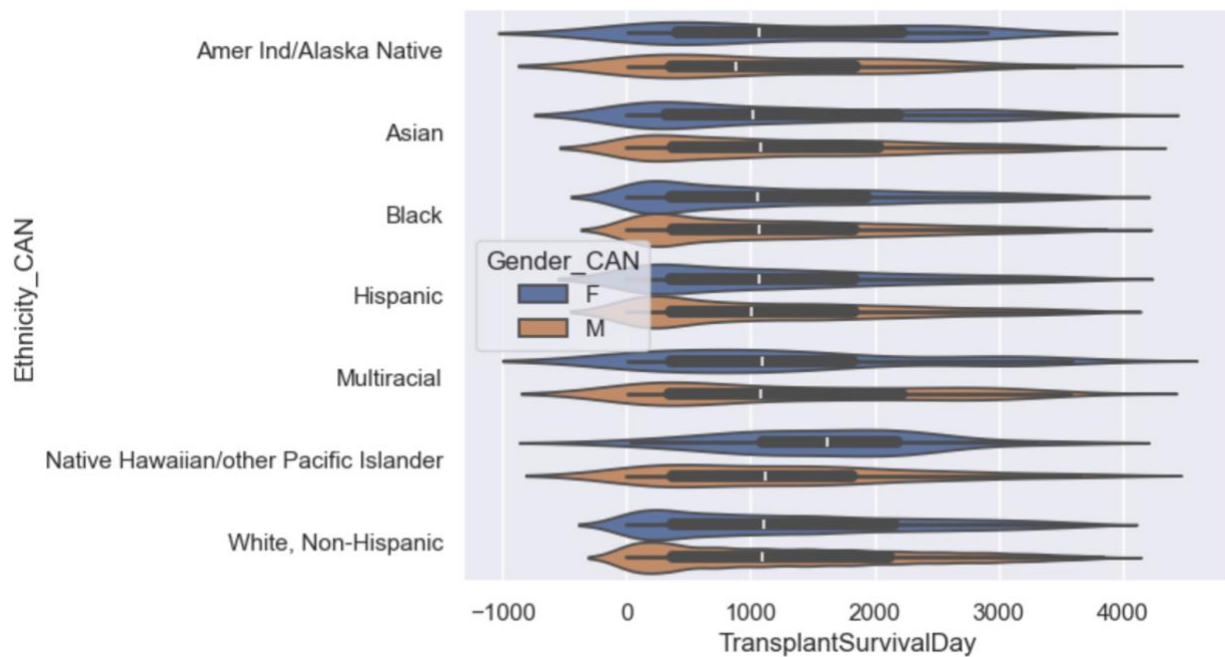


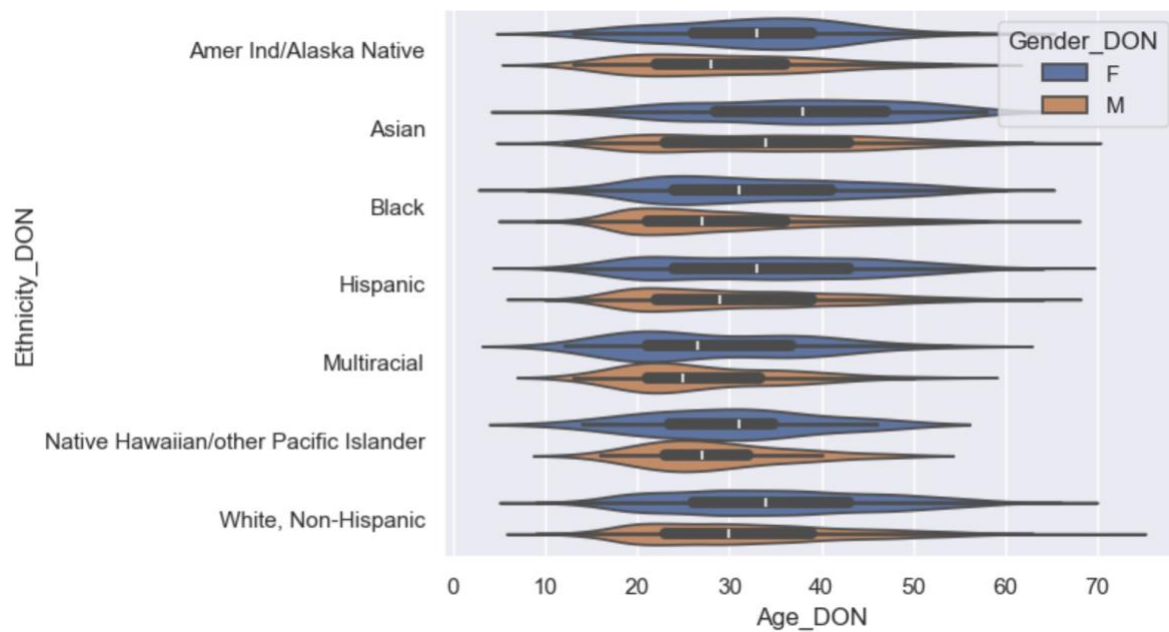
Histogram Plot of Ischemic Time Hour(s)



Histogram Plot of Distance From Donor Hospital To TX Center







REFERENCES

[1] <https://optn.transplant.hrsa.gov/news/first-heart-transplant-performed-35-years-ago/>